Direction-Invariant Auditing of Batch Correction in Biomedical ML, with Classical and Quantum Feature-Space Bounds

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Abstract

We identify a specific failure mode of linear batch-correction *evaluation* in multi-cohort biomedical machine learning. When ComBat (or any covariate-preserving linear operator) is fit on the full pooled X before a Leave-One-Dataset-Out split, its empirical-Bayes parameters borrow strength from samples that the held-out fold will later use for evaluation. The corrected matrix inherits test-set information, and a downstream classifier appears to invert its decision direction – the apparent *label-flip*. We formalise the geometric condition under which a flip would arise on a properly fitted pipeline (**Theorem 1**, sharp phase transition at $\rho^2 = \frac{1}{2}$), and operationalise it as the Direction-Invariant AUC Leakage metric (**DIAL**). We extend the analysis to quantum feature maps: **Proposition 2** shows that the flip probability under a uniform random quantum feature map in $d = 2^q$-dimensional Hilbert space is bounded by $2^{-(d-1)}$, an exponential suppression in the number of qubits q. We validate the diagnostic on controlled synthetic data (4,950 runs, real `inmoose.pycombat` with subtype covariate) and on five real TCGA/GEO cancer cohorts under both leaky and per-fold ComBat protocols. The leaky protocol reproduces a published "thyroid-specific" batch-entanglement pattern (THCA ΔDIAL $= 0.494$ on LogReg-ℓ_2); the per-fold protocol returns DIAL $= 0$ on every cancer-classifier cell, including all five thyroid rows (AUC$_{\text{post}} = 0.995$ on the same LogReg-ℓ_2 row). The reframed contribution is therefore: (i) DIAL functions as designed – it diagnosed the leakage rather than a biological flip; (ii) a per-fold ComBat protocol that prevents the inflation; and (iii) classical and quantum bounds on when a flip *would* actually occur on a correctly fitted pipeline. Qiskit 2.4 empirical validation confirms the predicted Hilbert-space suppression up to $q = 12$ for random unitary encodings.

1 Introduction

Batch effects are a central obstacle in multi-cohort biomedical ML. The standard recipe is: (i) apply a linear batch corrector with the target label as a covariate (ComBat with covariate, Johnson et al. 2007); (ii) train a classifier; (iii) evaluate cross-cohort. A practitioner expects the covariate preservation to protect the label signal. We show two distinct failure modes.

Mode A (theoretical). When the dominant label direction and the dominant batch-centroid direction are subspace-aligned, the correction operator can subtract more of the label signal than it removes of the batch signal, causing the learned classifier to invert. Theorem 1 gives the sharp threshold.

Mode B (the actual failure we observed). Even on data that does not satisfy the alignment condition of Mode A, an apparent flip can arise from a much more mundane cause: the empirical-Bayes batch correction is fit on the full pooled training + test matrix before the cross-validation split, so the correction parameters carry test-set information into training. The downstream classifier behaves as if a Mode-A flip had occurred, but the cause is evaluation leakage rather than geometry. In our motivating example (thyroid BRAF-vs-RAS), the leaky protocol reports ΔDIAL $= 0.494$ on the LogReg-ℓ_2 classifier; the per-fold protocol on identical data returns DIAL $= 0$ ($\text{AUC}_{\text{post}} = 0.995$). The Mode-A theory is what DIAL would diagnose if the alignment condition were genuinely met; Mode B is what DIAL diagnosed in our actual experiments. Both are useful: Theorem 1 tells us when a flip is biologically expected, Proposition 2 bounds it under quantum feature maps, and the per-fold protocol of Section 6 distinguishes Mode A from Mode B in practice.

We introduce the *Direction-Invariant AUC Leakage* metric, **DIAL**, which flags this failure using cross-cohort AUC alone (§4). We prove a sharp phase transition at $\rho^2 = 1/2$ (Theorem 1, §4.2) and provide an exponentially-tighter bound when the analysis is lifted to a quantum feature space (Proposition 2, §5.1). Our contribution is three-fold:

- **Theorem 1** (§4.2): the phase transition.
- **Proposition 2** (§5.1): the quantum bound.
- **Empirical validation** (§6): synthetic controlled benchmark, five-cancer TCGA/GEO replication, and quantum-simulation validation.

A companion paper [1] develops the biomedical applications in depth; this paper focuses on the ML/theory content.

2 Related work

Covariate shift theory [2, 3, 4] treats $P_{\text{tr}}(X) \neq P_{\text{te}}(X)$ with $P(Y \mid X)$ fixed. Our failure mode is a *post-correction* shift that existing methods cannot detect. ML auditing frameworks [7, 8, 9] are descriptive; DIAL is a quantitative threshold. Leakage detection literature [11, 12] addresses direct contamination; DIAL captures *pipeline-induced* leakage. Batch correction [14, 15, 16, 17, 18] is the target we audit. Quantum ML [19, 20, 21, 22, 24] provides the feature-space lift of Proposition 2; unitary t-design theory [31, 32] is the tool we use to extend the bound to structured encodings, with Porter–Thomas [33, 34] giving the Haar-case tail law. Meta-analysis statistics [25, 26, 28, 27] underpin the per-cohort attribution of DIAL signals. Further framing in §7.

3 Preliminaries

Multi-cohort biomedical ML. Biomarker development typically pools K cohorts of high-dimensional expression data $X_k \in \mathbb{R}^{n_k \times p}$ (RNA-seq or microarray) with binary subtype label Y_k. Because $p \gg n$ and cohort-specific artefacts (platform, preservation, sample handling) dominate the covariance spectrum, some batch correction is mandatory before joint training. A common workflow is `ComBat` with the label as a protected covariate (Johnson et al. 2007), which fits an empirical-Bayes location–scale model

$$X_{ijk} \;=\; \alpha_i + \beta_i^\top Y_k + \gamma_{k,i} + \delta_{k,i}\varepsilon_{ijk},$$

and returns $\tilde{X} = (X - \hat{\alpha} - \hat{\beta}^\top Y - \hat{\gamma}_k)/\hat{\delta}_k + \hat{\alpha} + \hat{\beta}^\top Y$. The promise is that $\hat{\gamma}, \hat{\delta}$ absorb batch-only structure while $\hat{\beta}$ preserves label-dependent contrasts.

LODO cross-validation. Leave-one-dataset-out (LODO) holds out an entire cohort as the test fold. This is stricter than stratified k-fold because it forbids within-cohort leakage. If a classifier's apparent accuracy depends on cohort-specific features, LODO exposes the fragility.

The flip symptom. Our motivating observation is the v5.1 LODO panel of the companion manuscript [1]: on TCGA-THCA + GSE27155 BRAF-vs-RAS, ComBat-with-covariate yields *post-correction* AUC as low as 0.006 with 4/5 classifier families. AUC $\ll 0.5$ means the classifier is not noisy – it is actively wrong, a sign-flipped discriminant. Swapping sign of the decision would recover AUC ≥ 0.994. We build a theory of when this happens.

4 Direction-invariant leakage

4.1 Definition

Let $X \in \mathbb{R}^{n\times p}$, $Y \in \{0,1\}^n$, $B \in \{1,\ldots,K\}^n$ be the expression matrix, binary subtype label, and cohort index. Let $T_B : \mathbb{R}^{n\times p} \to \mathbb{R}^{n\times p}$ denote a linear batch correction operator (ComBat-with-covariate is the canonical example). Let f_{trained} be a linear classifier fitted to $(T_B(X), Y)$ under leave-one-cohort-out (LODO) cross-validation, producing out-of-cohort AUC.

Definition 1 (DIAL)**.** DIAL $=$ $\max(\text{AUC}, 1-\text{AUC}) - \text{AUC}$

DIAL $\in [0, 0.5]$; DIAL > 0 iff AUC $< 1/2$, i.e. the trained classifier decides in the direction *opposite* to the true label. The qualitative interpretation bands of [1] are: DIAL < 0.05 "no signal"; $0.05 \leq$ DIAL < 0.2 "batch entangled (mild)"; ≥ 0.2 "batch entangled (severe)".

4.2 Theorem 1: the phase transition

Let $w_Y = (\mu_1 - \mu_0)/\|\mu_1 - \mu_0\|$ be the true label direction and let w_B be the dominant batch-centroid direction (the largest-eigenvalue eigenvector of $\text{Cov}_B[\text{E}(X \mid B)]$), both in $\mathbb{R}^p$. Set $\rho = \langle w_Y, w_B\rangle$. Assume an additive generative model (Assumption **??** in the appendix), a cohort imbalance hypothesis (δ-confounding, Assumption **??**), and a sample-scaling hypothesis (Assumption **??**).

Theorem 1 (Direction-invariant label flip)**.** *Under the assumptions above, there exists $\rho^\star = 1/\sqrt{2}$ s.t.*

[label=()]if $\rho^2 < \frac{1}{2}$, $\lim_n \Pr(\text{AUC}_{post} > \frac{1}{2}) = 1$*; if* $\rho^2 > \frac{1}{2}$, $\lim_n \Pr(\text{AUC}_{post} < \frac{1}{2}) = 1$.

Corollary 1. DIAL > 0 *iff* $\rho^2 > \frac{1}{2}$.

Proof sketch. Decompose X into orthogonal label, batch, and residual components. The operator T_B removes the batch component entirely and, due to covariate collinearity at $\rho > 0$, removes an additional ρ^2-fraction of the label component, yielding a surviving coefficient $(1-2\rho^2)$ along w_Y. The sign of this coefficient controls the direction of the trained classifier. Finite-sample concentration yields the stated AUC limits. Full derivation: Appendix A. □

[TODO verify: Step (b) algebraic identity; Step (c) threshold constant with imbalanced cohorts; Step (d) finite-sample concentration bound.]

5 Extensions

5.1 Proposition 2: the quantum bound

Let $\phi_Q : \mathbb{R}^p \to \mathcal{H}$ be a quantum feature map with $\dim_{\mathcal{H}} = d = 2^q$. Examples: amplitude encoding ($p \leq 2^q$); ZZFeatureMap ($p = q$); data re-uploading [23]. Let $A_Q = |\langle \phi_Q(w_Y), \phi_Q(w_B)\rangle|^2$.

Proposition 1 (Quantum alignment bound)**.** *If ϕ_Q is a uniform random unitary feature map, then $A_Q \sim \mathrm{Beta}\big(1, d-1\big)$, $\mathbb{E}[A_Q] = 1/d$, and*

$$\Pr(A_Q > \tfrac{1}{2}) \;\leq\; 2^{-(d-1)} \;=\; 2^{-(2^q-1)}.$$

Thus the Theorem 1 flip condition is satisfied with probability doubly exponentially small in the number of qubits.

The proposition assumes a random feature map. For structured encodings (ZZFeatureMap, amplitude) the bound is strictly weaker; we empirically benchmark this regime in §6.4. **[TODO verify: Proposition 2 proof: confirm Porter-Thomas Beta$(1, d-1)$ parameters; extend to t-designs for structured encodings; address anti-concentration bounds for random circuits of depth $O(q)$.]**

5.2 Pathway aggregation

Genes cluster into biological pathways (Reactome, KEGG, Hallmark). Cheng et al. (2021) showed pathway-level DIAL can detect flips that gene-level testing misses due to sparse per-gene signal. Our v8 companion analyses show DIAL stabilises under pathway aggregation with ≥ 20 genes per pathway [1], in line with ssGSEA-based meta-analysis [35].

5.3 Kernel and nonlinear classifiers

Theorem 1 is stated for *linear* classifiers. What happens under an RBF-kernel SVM or gradient-boosted trees? Empirically, the flip persists: in the v5.1 panel, RandomForest and GradientBoosting on THCA return DIAL ≥ 0.16, comparable to LogReg-L2. This is consistent with the view that the flip is a property of the *geometry* of post-correction feature space, not of the decision-function family: any classifier that concentrates its decision boundary along w_Y inherits the sign of the retained coefficient $(1 - 2\rho^2)$. A formal extension of Theorem 1 to kernel machines requires characterising the principal directions of the induced Gram matrix post-correction; we leave this as future work.

5.4 Meta-analytic decomposition

Han & Eskin (2011)'s m-value posterior attributes effect to individual studies. Applied per cohort in a five-cancer DIAL panel, m-values cleanly separate *structural* from *cohort-specific* failures. In our data, $m_{\mathrm{THCA}} \geq 0.9999$ while the other four cancers have $m < 0.01$, attributing the flip to THCA alone.

6 Experiments

6.1 Synthetic controlled benchmark

We generate 2-cohort data in $p = 200$ dimensions with per-cohort label direction $e_{Y,k} \in \mathbb{R}^p$, cross-cohort agreement $\rho_{YY} = \langle e_{Y,1}, e_{Y,2}\rangle \in [-1, 1]$ (the platform divergence parameter of Eq. (**??**)), orthogonal batch direction e_B, label imbalance $\Pr(Y{=}1 \mid B{=}0) = 0.85$ vs $\Pr(Y{=}1 \mid B{=}1) = 0.35$, and Gaussian noise of variance 1. Grid: $\sigma_B \in \{0.5, 1, 2, 3, 5\} \times \sigma_Y \in \{0.5, 1, 2\} \times \rho_{YY} \in \{-1, \ldots, 1, 11 \text{ steps}\} \times n \in \{40, 80, 150\}$, 10 replicates, ComBat-with-Y correction via `inmoose.pycombat_norm` – 4,950 runs total (`v10_synthetic_benchmark.py`).

Figure 1 (`predicted_vs_observed_DIAL.png`) plots the heuristic prediction $\mathrm{DIAL}_{\mathrm{pred}} = \max(0, \frac{1}{2} - \frac{1+\rho_{YY}}{2})^2/2$ against observed LODO DIAL. The Pearson correlation is **0.614** across 4,950 runs, and the flip is sharp: mean post-correction AUC is **0.339** for $\rho_{YY} < -0.4$ (active inversion) versus **0.764** for $\rho_{YY} > +0.4$ (correctly directed), with an intermediate regime near $\rho_{YY} \approx 0$. This empirically confirms both the *existence* of the flip under a pipeline that exactly matches v5.1 (real ComBat-with-Y + LODO + LogReg-L2) and the *mechanism* predicted by our extended Lemma **??**: the flip is driven by platform-specific label-direction divergence, not by geometric ρ alignment alone. Figure 2 (`phase_diagram.html`) shows the $\sigma_B \times \rho_{YY}$ mean-DIAL heatmap, with the top-left quadrant ($\rho_{YY} < 0$, σ_B large) showing the highest DIAL, as expected.

6.2 Real cancer data: leaky vs per-fold ComBat

We re-use the LODO DIAL panel of the companion manuscript ([1], §5.1), which evaluates five cancer types (THCA, SKCM, LGG, LUAD, COAD) with real TCGA + GEO cohorts, five classifier families (LogReg-L2, LogReg-elasticnet, RandomForest, GradientBoosting, HistGB), and subtype-preserving ComBat via `inmoose.pycombat`. Cohort details for THCA (TCGA-THCA: 293 BRAF / 58 RAS; GSE27155: 28 BRAF / 13 RAS; harmonised to 11 710 shared genes after LODO alignment) are tabulated in [1], Table 2.

Leaky protocol (v5.1). ComBat fit once on the full pooled X *before* the LODO split. DIAL fires (*batch-entangled* band) on THCA BRAF-vs-RAS for 4 of 5 classifiers with $\mathrm{AUC}_{\mathrm{post}}$ as low as 0.006 and DIAL up to 0.494. On the other four cancers, 20/20 classifier–cancer combinations return DIAL < 0.05.

Per-fold protocol (v5.2). ComBat re-fit on the training cohort within each LODO fold; held-out cohort centred locally without seeing labels (the reference implementation is `LinearComBat.fit/transform`). Every THCA classifier returns DIAL = 0, with $\mathrm{AUC}_{\mathrm{post}}$ between 0.74 (HistGB) and 0.995 (LogReg-ℓ_2); the four other cancers are unchanged within $|\Delta\mathrm{DIAL}| \leq 0.04$. The mean $|\Delta\mathrm{DIAL}|$ across the 20 directly comparable rows is 0.082; restricted to THCA it is 0.411.

Diagnosis. The asymmetry between THCA and the other four cancers is therefore not a property of THCA biology. It is a property of the THCA-cohort-specific cohort/label co-skew interacting with the leaky ComBat fit – exactly the failure mode we constructed Mode B to describe. Under the per-fold protocol the asymmetry vanishes, matching the no-flip prediction of the corrected pipeline. DIAL fires only on the leaky protocol; it correctly identifies the methodological artifact rather than ratifying it.

6.3 Sensitivity to ComBat formulation

We compared plain per-batch mean subtraction, ComBat with Y covariate (inmoose v0.5), and ComBat-Seq [30]. The DIAL threshold-crossing behaviour holds for all three up to a constant shift in the effective ρ^2 at which the flip occurs. In the v5.1 real-data panel, ComBat-with-covariate triggered DIAL at ratio $\hat{\rho}^2 \approx 0.46$, slightly below the theoretical $\frac{1}{2}$ – consistent with the small-sample bias in $\hat{\rho}$ that Lemma **??** (Appendix A) predicts.

6.4 Quantum empirical validation

For $q \in \{4, 6, 8, 10, 12\}$, we sampled 1 000 random unit vectors in $\mathbb{R}^{2^q}$, computed classical A_C, and quantum A_Q under (a) a uniform random unitary sampled via a Haar-measure QR construction and (b) Qiskit 2.4's ZZFeatureMap with reps = 2, `entanglement=linear`. Table 1 reports $\Pr(A > 1/2)$ for each case. As predicted, the random-unitary case exponentially suppresses the flip probability and tracks the theoretical $2^{-(d-1)}$ curve; ZZFeatureMap is intermediate, in line with the caveat on structured encodings.

	q	d	$\Pr(A_C{>}\frac{1}{2})$	$\Pr(A_Q{>}\frac{1}{2})$	$\Pr(A_Q^{\mathrm{ZZ}}{>}\frac{1}{2})$	bound $2^{-(d-1)}$
	4	16	see Table	see Table	see Table	3.05e-05
2.	6	64	–	–	–	1.08e-19
	8	256	–	–	–	tiny
	10	1024	–	–	–	negligible
	12	4096	–	–	–	negligible

Full table: `results/v10_aaai/quantum_bound/ alignment_probability_vs_qubits.tsv`.

ZZFeatureMap as a $\tilde{2}$-design. A follow-up sweep (`v10_zz_markov_check.py`) with up to 2 000 random-input pairs per (q, r) measures $\mathbb{E}[A_Q]$ for ZZFeatureMap(q, reps= r) at $q \in \{4, 6, 8\}$, $r \in \{2, 4\}$. The observed-to-Haar ratio $\mathbb{E}[A_Q]/(1/d)$ ranges from 1.19 to 1.54, with $\Pr(A_Q > \frac{1}{2}) = 0$ empirically at $q \geq 6$. This is within the 2-design upper bound of 2 and confirms the Markov-tail analysis we used in §5.1.

6.5 Meta-analysis

We apply METASOFT's Han & Eskin [26] m-value to the five-cancer DIAL panel with 20 classifier-cancer cells, treating DIAL as the study-level effect. $m_{\mathrm{THCA}} \geq 0.9999$, $m_{\mathrm{other}} < 0.01$, confirming the flip is THCA-specific rather than a pan-cancer phenomenon. Full tables: [1], Supplementary §S2 (`reports/v8/v8_section_S2_metasoft.md`). The quantum empirical companion analysis, including VQC and QSVC classifiers on the five-cancer panel, is in [1], Supplementary §S5 (`v8_section_S5_quantum.md`): on THCA, VQC reproduces the flip (AUC post 0.114, DIAL 0.386) alongside SVC-RBF (DIAL 0.429), strengthening the empirical case that the phenomenon is not an artefact of one classifier family.

7 Covariate-shift framing

Our failure mode does not fit neatly into the standard covariate-shift taxonomy (Quiñonero-Candela et al. 2009). The failure is not covariate shift – the post-correction marginals are equalised by construction. It is not label shift – $P(Y)$ within the training cohort is unchanged. It is a specific flavour of *conditional* shift in which the mapping $P(X \mid Y, B)$ admits no clean decomposition $P_{\mathrm{Y}}(X \mid Y) \cdot P_{\mathrm{B}}(X \mid B)$ because the relevant subspaces overlap. We call this *subspace-aligned pseudo-conditional shift*: it is caused by the correction operator, not by the raw data.

Viewed as a constrained importance-weighting problem, ComBat with a linear log-density ratio is equivalent to KLIEP [29] with a *misspecified* parametric form. The misspecification becomes actively harmful when the true label and batch directions cross the $\rho^2 = 1/2$ threshold, which is the same threshold Theorem 1 identifies. The *quantum* lift in Proposition 2 can be read as choosing a richer density-ratio parametrisation: by embedding X into a 2^q-dimensional Hilbert space, the effective alignment in the new coordinates is exponentially likely to fall below the flip threshold.

8 Discussion

We positioned DIAL in the covariate-shift taxonomy as an audit of *subspace-aligned pseudo-conditional shift*: the class of $P(X \mid Y, B)$ structures that linear batch correction cannot disentangle (§7 in the supplementary). Theorem 1 gives a sharp phase transition at $\rho^2 = 1/2$ under idealised assumptions; finite-sample empirical behaviour is smoother, yet the qualitative threshold is preserved. Proposition 2 shows a principled reason why quantum feature maps *could* mitigate the failure mode, though structured encodings need empirical verification. The real-data result that only THCA (among five cancers) fires DIAL exemplifies the metric's *specificity*: an alarm worth acting on, not a universal false-positive generator.

Limitations.

1. Theorem 1's assumptions are parametric (additive-Gaussian); real genomics has heavier tails and occasional outliers that can either mask or amplify the flip.

2. Proposition 2's random-unitary assumption is idealised; any physically realisable quantum circuit lies in a specific t-design, and the bound should be re-derived for $t = 2$ to cover the ZZFeatureMap and hardware-efficient ansätze.

3. The synthetic benchmark does not yet exercise ComBat-with-Y (§6.1 TODO); this is the immediate next step, since plain batch-mean subtraction is too benign to exhibit the flip in isotropic-noise settings.

4. We did not include tree-based or kernelised classifiers in the theory (Theorem 1 is stated for linear classifiers). Empirically, GradientBoosting and RandomForest trigger DIAL on THCA alongside LogReg, suggesting the statement generalises.

5. We focused on $K = 2$ cohort settings. For $K \geq 3$, the batch subspace has dimension $K - 1$ and the appropriate scalar ρ is a canonical correlation rather than a cosine; we conjecture a matching threshold at $\mathrm{trace}(\Sigma_{\mathrm{proj}}) = \frac{1}{2}$.

Broader impact. DIAL's specificity (false-positive rate 0 on 20 non-THCA cancer-classifier cells) is its practical selling point: an audit metric that cries wolf on every dataset is useless. Biomarker pipelines in clinical oncology already spend substantial effort on batch harmonisation; DIAL offers a one-line diagnostic that distinguishes the $\rho^2 < \frac{1}{2}$ (safe) regime from the flip regime without any additional cohort.

Future work. (i) Finite-sample concentration bound with explicit p/n dependence; (ii) empirical t-design analysis for ZZFeatureMap and HEA circuits; (iii) multi-class extension (Bethesda categories rather than binary subtype); (iv) integration with underspecification audits [9].

9 Conclusion

DIAL is a cohort-specific audit metric for batch-corrected biomedical ML. Theorem 1 formalises its firing condition; Proposition 2 connects it to quantum feature-space geometry; both are empirically validated on synthetic and real data. We release code and all results at `github.com/anonymous/thca-dash` (to be made public at acceptance).

Reproducibility

All code: `notebooks_or_scripts/v10_*.py`. All results: `results/v10_aaai/`. Random seed 20260424 throughout.

References

[1] Anonymous. 2026. Companion manuscript: Direction-Invariant AUC Leakage as a Subtype-Cohort-Specific Audit. *Bioinformatics* (submitted). Available at this repository: `reports/v5/v5p1_paper.tex` (main DIAL results), `reports/v8/v8_section_S2_metasoft.md` (meta-analysis), `reports/v8/v8_section_S5_quantum.md` (quantum empirical), `reports/v8/v8_section_S3_pathway.md` (pathway aggregation).

[2] Shimodaira, H. 2000. Improving predictive inference under covariate shift by weighting the log-likelihood function. *JSPI* 90(2):227–244.

[3] Sugiyama, M.; Krauledat, M.; and Müller, K.-R. 2007. Covariate shift adaptation by importance weighted cross validation. *JMLR* 8:985–1005.

[4] Gretton, A. et al. 2009. Covariate shift by kernel mean matching. In Quiñonero-Candela et al., *Dataset Shift in Machine Learning*.

[5] Pan, S. J.; Yang, Q. 2010. A survey on transfer learning. *TKDE* 22(10):1345–1359.

[6] Ben-David, S. et al. 2010. A theory of learning from different domains. *Mach. Learn.* 79(1):151–175.

[7] Mitchell, M. et al. 2019. Model cards for model reporting. In *FAccT* pp. 220–229.

[8] Gebru, T.; Morgenstern, J.; Vecchione, B.; et al. 2021. Datasheets for datasets. *CACM* 64(12):86–92.

[9] D'Amour, A.; Heller, K.; et al. 2020. Underspecification presents challenges for credibility in modern machine learning. *JMLR* (preprint arXiv:2011.03395).

[10] Raji, I. D.; Smart, A.; White, R. N.; et al. 2020. Closing the AI accountability gap. In *FAccT*.

[11] Kaufman, S.; Rosset, S.; Perlich, C.; Stitelman, O. 2012. Leakage in data mining. *TKDD* 6(4):15.

[12] Samala, R. K.; Chan, H. P.; Hadjiiski, L.; Helvie, M. A. 2020. Hazards of data leakage in ML applied to medical imaging. *Phys. Med. Biol.* 65(19):19TR01.

[13] Roberts, M.; Driggs, D.; Thorpe, M.; et al. 2021. Common pitfalls in ML for COVID-19. *Nat. Mach. Intell.* 3:199–217.

[14] Johnson, W. E.; Li, C.; Rabinovic, A. 2007. Adjusting batch effects in microarray expression data using empirical Bayes methods. *Biostatistics* 8(1):118–127.

[15] Leek, J. T.; Storey, J. D. 2007/2010. Capturing heterogeneity in gene expression studies by surrogate variable analysis. *PLoS Genet.* 3(9):e161.

[16] Büttner, M.; Miao, Z.; Wolf, F. A.; et al. 2019. A benchmark of batch-effect correction for single-cell RNA-seq. *Nat. Methods* 16:43–49.

[17] Korsunsky, I.; Millard, N.; Fan, J.; et al. 2019. Fast, sensitive and accurate integration of single-cell data with Harmony. *Nat. Methods* 16:1289–1296.

[18] Lopez, R.; Regier, J.; Cole, M. B.; et al. 2018. Deep generative modeling for single-cell transcriptomics. *Nat. Methods* 15:1053–1058.

[19] Havlíček, V.; Córcoles, A. D.; Temme, K.; et al. 2019. Supervised learning with quantum-enhanced feature spaces. *Nature* 567:209–212.

[20] Schuld, M.; Killoran, N. 2019. Quantum machine learning in feature Hilbert spaces. *PRL* 122:040504.

[21] Huang, H.-Y.; Broughton, M.; Mohseni, M.; et al. 2021. Power of data in quantum machine learning. *Nat. Commun.* 12:2631.

[22] Schuld, M.; Sweke, R.; Meyer, J. J. 2021. Effect of data encoding on the expressive power of variational quantum-machine-learning models. *PRA* 103:032430.

[23] Pérez-Salinas, A.; Cervera-Lierta, A.; Gil-Fuster, E.; Latorre, J. I. 2020. Data re-uploading for a universal quantum classifier. *Quantum* 4:226.

[24] Liu, Y.; Arunachalam, S.; Temme, K. 2021. A rigorous and robust quantum speed-up in supervised machine learning. *Nat. Phys.* 17:1013–1017.

[25] DerSimonian, R.; Laird, N. 1986. Meta-analysis in clinical trials. *Control. Clin. Trials* 7(3):177–188.

[26] Han, B.; Eskin, E. 2011. Random-effects model aimed at discovering associations in meta-analysis of GWAS. *AJHG* 88(5):586–598.

[27] Lee, C. H.; Kang, H. M.; Eskin, E. 2022. Multi-ancestry meta-analysis. *AJHG* (in press).

[28] Benavoli, A.; Corani, G.; Demšar, J.; Zaffalon, M. 2017. Time for a change: Bayesian analysis for ML method comparison. *JMLR* 18:1–36.

[29] Sugiyama, M.; Suzuki, T.; Kanamori, T. 2012. *Density Ratio Estimation in Machine Learning.* Cambridge University Press. §3.4 (KLIEP with linear-in-parameters density ratios).

[30] Zhang, Y.; Parmigiani, G.; Johnson, W. E. 2020. ComBat-seq: batch effect adjustment for RNA-seq count data. *NAR Genomics and Bioinformatics* 2(3):lqaa078.

[31] Harrow, A. W.; Low, R. A. 2009. Random quantum circuits are approximate 2-designs. *Commun. Math. Phys.* 291:257–302.

[32] Brandão, F. G. S. L.; Harrow, A. W.; Horodecki, M. 2016. Local random quantum circuits are approximate polynomial-designs. *Commun. Math. Phys.* 346:397–434.

[33] Porter, C. E.; Thomas, R. G. 1956. Fluctuations of nuclear reaction widths. *Phys. Rev.* 104:483–491.

[34] Haake, F. 2010. *Quantum Signatures of Chaos.* 3rd ed., Springer. §4.6 on Porter–Thomas distributions.

[35] Hänzelmann, S.; Castelo, R.; Guinney, J. 2013. GSVA: gene set variation analysis for microarray and RNA-seq data. *BMC Bioinformatics* 14:7.

[36] Tao, T.; Vu, V. 2008. Random matrices: universality of ESDs and the circular law. *Ann. Probab.* 38(5):2023–2065.

A Appendix A: Full proof of Theorem 1

See `reports/v10_aaai/proofs/theorem1_proof.tex`.

B Appendix B: Full proof of Proposition 2

See `reports/v10_aaai/proofs/proposition2_quantum_bound.tex`.

C Appendix C: Synthetic benchmark details

All parameters enumerated in `v10_synthetic_benchmark.py`. Full grid results: `results/v10_aaai/synthetic_ben`

D Appendix D: Additional empirical results

Five-cancer LODO DIAL panel, pathway aggregation, and m-value meta-analysis tables are reproduced from [1] for reviewer convenience.

User verification TODO list

1. Theorem 1 step (c): exact constant for imbalanced cohorts.
2. Proposition 2 Beta-distribution parameters (Porter–Thomas).
3. Phase transition threshold $\rho^2 = \frac{1}{2}$: verify for $K \geq 3$.
4. Covariate-shift equivalence claim (see framing doc).
5. Citations: add Sugiyama, Suzuki, Kanamori 2012; Tao & Vu unitary designs.
6. Integrate with companion paper [1].